

Question – 1 :-

1. Create a new GitHub repository.

- Clone the repository to your local machine using SSH (generate an SSH key if needed, add the public key to your GitHub account).
- Create a new branch named after your username (e.g., `Tutedude`).
- Add your Flask project files to this branch.
- Commit the changes and merge the branch into the `main` branch.

Solution :-

```
ubuntu@ip-172-31-22-169: ~/vcs_lakshman
hp@LAPTOP-80EBCM7D MINGW64 ~
$ ssh -i Downloads/docker-key.pem ubuntu@44.222.206.165
welcome to ubuntu 26.04 LTS (GNU/Linux 7.0.0-1008-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Fri Jul 24 09:58:43 UTC 2026

System load:  0.14           Processes:            115
Usage of /:   30.0% of 18.25GB Users logged in:          0
Memory usage: 36%           IPv4 address for enx0: 172.31.22.169
Swap usage:   0%

* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

29 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Jul 19 10:05:19 2026 from 106.219.156.215
ubuntu@ip-172-31-22-169:~$ ssh-keygen -t ed25519 -C "lakshmanmadheshiya98@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
/home/ubuntu/.ssh/id_ed25519 already exists.
Overwrite (y/n)?
ubuntu@ip-172-31-22-169:~$ ssh-keygen -t ed25519 -C "lakshmanmadheshiya98@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
/home/ubuntu/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:RcQj9Ba/YwBAKCZPCD30wy48hdeqlaYEJuAjmRyhPHQ lakshmanmadheshiya98@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|o=OE.OO000+      |
|O+* ++ ..+00     |
|*@O...* ..=.    |
|.O+O + + O . .   |
|  = * S  +       |
|. B  . .         |
| o              |
|                |
+-----[SHA256]-----+
```

```

ubuntu@ip-172-31-22-169:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI3m1E5UARC12bzmPyGr/GqRc6X6Iq3ZZzJ7srsmQ2Fjg lakshmanmadheshiya98@gmail.com
ubuntu@ip-172-31-22-169:~$ ssh -T git@github.com
Hi laksh1001! You've successfully authenticated, but GitHub does not provide shell access.
ubuntu@ip-172-31-22-169:~$ git clone git@github.com:laksh1001/your-repo-name.git
Cloning into 'your-repo-name'...
ERROR: Repository not found.
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
ubuntu@ip-172-31-22-169:~$ git remote set-url origin git@github.com:laksh1001/vcs_lakshman.git
fatal: not a git repository (or any of the parent directories): .git
ubuntu@ip-172-31-22-169:~$ git clone git@github.com:laksh1001/vcs_lakshman.git
Cloning into 'vcs_lakshman'...
warning: You appear to have cloned an empty repository.
ubuntu@ip-172-31-22-169:~$ cd vcs_lakshman/
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ echo "# VCS Lakshman Project" > README.md
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add README.md
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git commit -m "Initial commit"
[main (root-commit) 3498d2a] Initial commit
  Committer: ubuntu <ubuntu@ip-172-31-22-169.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 1 insertion(+)
 create mode 100644 README.md
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 246 bytes | 246.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:laksh1001/vcs_lakshman.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
ubuntu@ip-172-31-22-169:~/vcs_lakshman$

```

Question – 2 :-

2. Create a new branch named `<your_name>_new` (e.g., `Tutedude_new`).
 - Update the content of the JSON file used for the `/api` route in this branch.
 - Merge the `<your_name>_new` branch into the `main` branch.
 - If there are conflicts during the merge, resolve them by accepting the changes from the `<your_name>_new` branch.
 - Add the resolved changes to the staging area, commit them, and push the updates to the remote repository.

Solution :-

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hp@LAPTOP-80EBCM7D MINGW64 ~

```
$ ssh -i Downloads/docker-key.pem ubuntu@44.222.206.165
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1008-aws x86_64)
```

```
* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Sat Jul 25 04:49:22 UTC 2026

```
System load:  0.03      Processes:      115
Usage of /:   30.0% of 18.25GB   Users logged in:  0
Memory usage: 36%          IPv4 address for enX0: 172.31.22.169
Swap usage:   0%
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<https://ubuntu.com/aws/pro>

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Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

Last login: Fri Jul 24 09:58:44 2026 from 106.219.155.212

ubuntu@ip-172-31-22-169:~\$ cd vcs_lakshman/

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git checkout -b lakshman_new

Switched to a new branch 'lakshman_new'

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ ls

README.md

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ nano data.json

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git add .

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git commit -m "Update JSON file for /api route"

[lakshman_new 8e51d58] Update JSON file for /api route

1 file changed, 9 insertions(+)

create mode 100644 data.json

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git checkout main

Switched to branch 'main'

Your branch is up to date with 'origin/main'.

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git merge lakshman_new

Updating 3498d2a..8e51d58

Fast-forward

data.json | 9 ++++++++

1 file changed, 9 insertions(+)

create mode 100644 data.json

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git checkout --theirs .

Updated 0 paths from the index

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git add .

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git commit -m "Resolved conflicts by keeping lakshman_new changes"

On branch main

Your branch is ahead of 'origin/main' by 1 commit.

(use "git push" to publish your local commits)

nothing to commit, working tree clean

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ git push origin main

Enumerating objects: 4, done.

Counting objects: 100% (4/4), done.

Compressing objects: 100% (3/3), done.

Writing objects: 100% (3/3), 437 bytes | 437.00 KiB/s, done.

Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

To github.com:laksh1001/vcs_lakshman.git

3498d2a..8e51d58 main -> main

ubuntu@ip-172-31-22-169:~/vcs_lakshman\$ |

Question – 3 :-

3. Branch Creation:

- Create two branches: `master_1` and `master_2` from the `main` branch.
- **Feature Development in `master_1`:**
- In the `master_1` branch, create a **To-Do Page** in the frontend.
 - The page should contain a form with the following fields:
 - **Item Name**
 - **Item Description**
- **Backend API in `master_2`:**
- In the `master_2` branch, create a backend route named `/submittodoitem`.
- This route will:
 - Accept `itemName` and `itemDescription` via a POST request.
 - Store these details in a MongoDB database.
- **Merging Changes:**
- Merge the changes from both `master_1` and `master_2` into the `main` branch.

Solution :-

```
ubuntu@ip-172-31-22-169: ~/vcs_lakshman
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout -b master_1
Switched to a new branch 'master_1'
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout -b master_2
Switched to a new branch 'master_2'
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout master_1
Switched to branch 'master_1'
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ nano todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git commit -m "Add To-Do Page frontend with form fields"
[master_1 b05b0ba] Add To-Do Page frontend with form fields
1 file changed, 19 insertions(+)
 create mode 100644 todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout master_2
Switched to branch 'master_2'
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ nano server.js
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add server.js
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git commit -m "Add /submittodoitem backend POST route for MongoDB"
[master_2 3df6b01] Add /submittodoitem backend POST route for MongoDB
1 file changed, 25 insertions(+)
 create mode 100644 server.js
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git merge master_1
Updating 8e51d58..b05b0ba
Fast-forward
 todo.html | 19 ++++++
 1 file changed, 19 insertions(+)
 create mode 100644 todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git merge master_2
Merge made by the 'ort' strategy.
 server.js | 25 ++++++
 1 file changed, 25 insertions(+)
 create mode 100644 server.js
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 1.35 KiB | 1.35 MiB/s, done.
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To github.com:laksh1001/vcs_lakshman.git
 8e51d58..ef7e4d0  main -> main
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git push origin master_1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master_1' on GitHub by visiting:
remote:   https://github.com/laksh1001/vcs_lakshman/pull/new/master_1
remote:
To github.com:laksh1001/vcs_lakshman.git
 * [new branch]   master_1 -> master_1
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git push origin master_2
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master_2' on GitHub by visiting:
remote:   https://github.com/laksh1001/vcs_lakshman/pull/new/master_2
remote:
To github.com:laksh1001/vcs_lakshman.git
 * [new branch]   master_2 -> master_2
ubuntu@ip-172-31-22-169:~/vcs_lakshman$
```

Question – 4 :-

4. Enhancing the To-Do Form in `master_1`:

- In the `master_1` branch, add the following fields to the To-Do form:
 - **Item ID**
 - **Item UUID**
 - **Item Hash**
- **Committing in Sequence:**
- Add and commit each field separately in the following order:
 - **First commit:** Add **Item ID** field.

- **Second commit:** Add **Item UUID** field.
 - **Third commit:** Add **Item Hash** field.
- **Merging to `main`:**
- Merge the `master_1` branch into the `main` branch.
- **Git Reset and Commit Deletion:**
- In the `main` branch, use **Git Reset** to roll back to the commit where only the **Item ID** field was added.
- Use `git reset --soft` to ensure changes remain staged.
- Re-commit this state to the `main` branch.
- Merge this updated state to the `main` branch.
- **Rebasing Changes:**
- Rebase the updated changes in the `main` branch to the `master_1` branch.
- **Clarification:**
 - During rebasing, **preserve individual commits** to maintain the commit history for each change (i.e., do not squash commits).
 - Use `git rebase main master_1` to integrate changes from the `main` branch back into the `master_1` branch.

Solution :-

```

ubuntu@ip-172-31-22-169: ~/vcs_lakshman
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout master_1
Switched to branch 'master_1'
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ nano todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git commit -m "Add Item ID field"
[master_1 ee3300f] Add Item ID field
1 file changed, 3 insertions(+)
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ nano todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git commit -m "Add Item UUID field"
[master_1 19358c0] Add Item UUID field
1 file changed, 3 insertions(+)
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ nano todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git commit -m "Add Item Hash field"
[master_1 07c39d0] Add Item Hash field
1 file changed, 3 insertions(+)
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git merge master_1
Merge made by the 'ort' strategy.
todo.html | 9 ++++++++
1 file changed, 9 insertions(+)
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git reset --soft HEAD~2
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git commit -m "Update To-Do form state with staged changes"
[main f9466d4] Update To-Do form state with staged changes
2 files changed, 34 insertions(+)
create mode 100644 server.js
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git rebase main master_1
Auto-merging todo.html
CONFLICT (content): Merge conflict in todo.html
error: could not apply ee3300f... Add Item ID field
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply ee3300f... # Add Item ID field

```

```

Could not apply ee3300f... # Add Item ID field
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout --ours todo.html
Updated 1 path from the index
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git rebase --continue
Auto-merging todo.html
CONFLICT (content): Merge conflict in todo.html
error: could not apply 19358c0... Add Item UUID field
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 19358c0... # Add Item UUID field
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout --ours todo.html
Updated 1 path from the index
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git rebase --continue
dropping 07c39d0cd81f65120fa67b7971b913af541147a6 Add Item Hash field -- patch contents already upstream
Successfully rebased and updated refs/heads/master_1.
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout --ours todo.html
Updated 0 paths from the index
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git add todo.html
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git rebase --continue
fatal: no rebase in progress
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout main
Switched to branch 'main'
Your branch and 'origin/main' have diverged,
and have 1 and 2 different commits each, respectively.
(Use "git pull" if you want to integrate the remote branch with yours)
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git push origin main --force
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 847 bytes | 847.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:laksh1001/vcs_lakshman.git
+ ef7e4d0...f9466d4 main -> main (forced update)
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git checkout master_1
Switched to branch 'master_1'
ubuntu@ip-172-31-22-169:~/vcs_lakshman$ git push origin master_1 --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:laksh1001/vcs_lakshman.git
b05b0ba..f9466d4 master_1 -> master_1
ubuntu@ip-172-31-22-169:~/vcs_lakshman$

```